



Tariffs and the Exchange Rate: Evidence from Twitter

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Abstract

This paper examines the conjecture that an increase in tariffs in a flexible exchange rate regime leads to the appreciation of the local currency. We focus on the reaction of the exchange rate market to tweets by U.S. President Donald Trump regarding possible tariff increases on Canadian and Mexican goods. The anticipation of trade restrictions leads to the appreciation of the U.S. dollar by 2.6 bps and 4.6 bps vis-à-vis the Canadian dollar and Mexican peso, respectively, within five minutes of the tweet, and the effect is statistically significant for up to 3 days after the tweet. These results suggest that the view that exchange rate appreciation may mitigate expenditure-switching toward local goods after a tariff increase is empirically plausible.

Keywords NAFTA · Commercial policy · Government communication · Expenditure-switching · High-frequency identification · Social media

JEL Classification F13 · F31

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1 Introduction

This paper examines the conjecture that an increase in tariffs on foreign goods in a flexible exchange rate regime leads to the appreciation of the local currency. The focus is on government communication in the form of tweets issued by U.S. President Donald Trump regarding actual and possible tariffs on Canadian and Mexican goods. The president has extensive statutory powers on international trade matters, and his tweets are informative and potentially consequential.¹ Canada and Mexico are, respectively, the second and third largest U.S. trade partners; and the effects of trade restrictions, even if just as a possibility or a threat, may be large and elicit a response in foreign exchange markets. The facts that the data are well-measured, relatively abundant, available at high frequencies, and concern countries that jointly account for one-third of U.S. trade means that we are able to identify the effects of trade restrictions on the exchange rate.

Our work is empirical in nature, but it sheds light on the theoretical literature that asks whether an increase in tariffs on foreign goods is expansionary. This is an important policy issue because countries (e.g., the U.S. during the presidency of Donald Trump) can engage in commercial policy in an attempt to influence their aggregate output and trade balance. However, if the local currency appreciates following the imposition of tariffs, then relative prices may be left unchanged and the intended expenditure-switching from foreign to local goods would not take place. A classical paper by Robert Mundell (Mundell 1961) suggests that in a flexible exchange rate regime, an increase in tariffs causes the appreciation of the local currency and, contrary to the possible intention of the policy, domestic output may fall because the improvement in the terms of trade increases savings via the Laursen–Metzler effect and reduces aggregate demand (see Laursen and Metzler 1950). Similar results are reported by Chan (1978), Eichengreen (1981), Krugman (1982), and Ostry (1991).

Recent work that studies the effects of commercial policy includes Barattieri et al. (2021), Erceg et al. (2018), and Caldara et al. (2021).² Barattieri et al. examine the effect of an increase in protectionism by a small open economy under a flexible exchange rate and find that expenditure-switching toward local goods is partly mitigated by the appreciation of the exchange rate so that the overall effect of the policy is contractionary. In contrast, Erceg et al. find that in a New Keynesian open-economy model, a combination of import tariffs and export subsidies may be expansionary. Caldara et al. find that uncertainty about higher future tariffs reduces investment and economic activity.

As the literature above indicates, a key issue is the behavior of the exchange rate following the increase in tariffs. A challenge to study this behavior empirically is

¹ For example, the Trump administration brought about the imposition of 25% and 10% duties on steel and aluminum imports from Canada and Mexico in June 2018 (see *The New York Times*, May 31, 2018).

² A parallel literature that studies the effects of trade sanctions was spurred by the Russian invasion of Ukraine in 2022. Theoretical models of the exchange rate response to sanctions are developed in Ghironi et al. (2022), Itskhoki and Mukhin (2022), and Lorenzoni and Werning (2022). Eichengreen et al. (2022) examine empirically the impact of sanctions on the exchange rate in the period from 1914 to 1945.



that changes in U.S. tariffs were relatively infrequent until recently, and when they occurred, they covered narrowly defined product categories. Hence, it may be hard to detect aggregate effects based on historical data and to identify a causal relation between tariff changes and exchange rate movements. Ostry and Rose (1992) and, more recently, Barattieri et al. (2021) and Furceri et al. (2019) use vector autoregressions to examine the effect of tariff changes on several macroeconomic variables, including the exchange rate. Jeanne and Son (2020) examine instances where the U.S. imposed tariffs on Chinese goods and find that U.S. tariffs lead to the appreciation of the U.S. dollar but that Chinese tariffs do not have a significant effect on the renminbi. Amiti et al. (2021) show that trade-war announcements between the U.S. and China led to declines in U.S. stock prices, expected productivity, and expected inflation.³ We complement their work by focusing on the exchange rate market's reaction to the possible tariff increases that would have taken place if the United States had withdrawn from the North American Free Trade Agreement (NAFTA) or if the negotiations for a new trade agreement would have failed. Trade with the other NAFTA members accounted in 2016 for 68%, 66%, and 29% of all international trade by Canada, Mexico, and the U.S., respectively.⁴ Hence, the aggregate implications of NAFTA's collapse are likely to be large and the foreign exchange market would react to the possibility of this outcome.⁵

The uncertainty regarding NAFTA was driven by the "America First" policy during the presidency of Donald Trump. Although the U.S. Congress has competency over trade policy, the powers of the president have been considerably expanded through the Trade Expansion Act of 1962 and the Trade Act of 1974. The United States Trade Representative is part of the Executive Office of the President and is the agency responsible for developing and recommending trade policy to the president, conducting trade negotiations, and coordinating trade policy with other government agencies. During the period considered, a key source of news regarding trade policy and, in particular, tariffs were public communications by the U.S. president, frequently in the form of tweets.

We quantify the causal effect on the bilateral exchange rates of the possible and actual imposition of tariffs on Canada and Mexico as gleaned from presidential tweets. Identification is based on a high-frequency strategy that computes the change in the nominal exchange rate between two points in a narrow time window around the Twitter feed. Other news could induce movements in the exchange rate during the day, and the identifying assumption is that the Twitter feed was the only source of new information arriving during the brief time interval around it. Our exchange rate data correspond to the nominal rate, but it is safe to assume that the price level is unchanged over the time window around the tweet, so that a high-frequency

³ Auray et al. (2021); Auray et al. (2022) show that the choice of tariffs in a trade war depends on the exchange rate regime. The incentive to impose tariffs is larger in a floating than in a fixed regime because the latter precludes the manipulation of the terms of trade. An important difference between NAFTA negotiations and the U.S.–China trade conflict is that China retaliated immediately and proportionally to U.S. tariffs, while Canada and Mexico's response was more muted and delayed. This means that retaliation is less likely to impinge on our empirical results.

⁴ Authors' calculations are based on data from the World Bank available at <https://wits.worldbank.org/>.

⁵ For an analysis of the macroeconomic effects of NAFTA's termination, see Steinberg (2020).



movement in the nominal exchange rate translates one-to-one into a high-frequency movement in the real exchange rate.

This high-frequency identification strategy is used by previous literature that studies the effect of monetary policy announcements on long-term interest rates (Cochrane and Piazzesi 2002; Gürkaynak et al. 2002), on the volatility of the stock returns of individual firms (Gorodnichenko and Weber 2016), and on expected inflation and output growth (Nakamura and Steinsson 2018). The strategy is also used by a recent literature that examines the effect of tweets on the Chinese and U.S. stock markets (Blanchard and Collins (2019)) and on the federal funds futures market (Camous and Matveev 2021; Bianchi et al. 2023).⁶ In the context of fiscal policy, the literature also includes (Romer and Romer 2010), who use government publications and presidential speeches to construct a time series of tax changes and quantify their effect on aggregate output, and Ramey (2011), who examines the effects of news regarding military spending as reported by *Business Week*. Breinlich et al. (2018) study the effect on stock returns of Brexit-related speeches by Theresa May. In the same spirit, our paper is concerned with the effects of government communication on economic variables and uses social media postings to shed light on the response of the exchange rate to changes in tariffs.

Our results show that the anticipation of trade restrictions leads to the appreciation of the U.S. dollar by 2.61 and 4.56 basis points (bps) with respect to the Canadian dollar and Mexican peso, respectively, within five minutes of the tweet. Forward rates up to five years ahead respond in a quantitatively similar manner. These figures are economically and statistically significant and robust to the estimated specification. Moreover, local projections show that the effects of presidential tweets persist for several days and are statistically significant for up to 3 days. Hence, government communication in the form of tweets can have effects as long lasting as, for example, press conferences following FOMC meetings (see Gorodnichenko et al. 2023). The finding that the effect is larger on the Mexican peso than on the Canadian dollar is likely due to the facts that 1) Canada could have continued to trade freely with the U.S. after NAFTA's termination under the previous Canada–U.S. Free Trade Agreement, 2) the aggregate effects of NAFTA's termination would have been larger for Mexico than for Canada (Steinberg 2020), and 3) the language in tweets concerning Mexico was considerably more aggressive than those concerning Canada and more likely to receive the attention of traders. In summary, our results suggest that the conjecture that exchange rate appreciation may mitigate expenditure-switching toward local goods after a tariff increase is empirically plausible.

This paper is organized as follows: Section 2 describes the data used, Sect. 3 presents case studies that document the effects of presidential tweets on exchange rates, Sect. 4 reports empirical results, Sect. 5 interprets our results through the lens of a structural model, and Sect. 6 outlines the policy implications of our analysis.

⁶ See also Gholampour and van Wincoop (2019), who use private tweets about the euro/U.S. dollar exchange rate to evaluate to what extent it is driven by fundamentals or by private information.



2 Data and the Role of the Presidency in U.S. Trade Policy

The raw data for this study consist of 336 tweets by Donald Trump, first as candidate and then as president, between January 1, 2016, and November 29, 2019. We selected these tweets from more than 40,000 tweets available at the publicly available repository, www.thetrumparchive.com, using the keywords: Canada, Mexico, Canadian, Mexican, NAFTA, USMCA, and CUSMA. We disregard re-tweets, which by definition contain no new information, and weekend tweets, when foreign exchange markets are closed.⁷ We consider tweets from the same thread jointly if they fall within the same identification window and as separate tweets otherwise. The identification window is six minutes long. The window starts one minute before and ends five minutes after the tweet publication. To verify that our identification strategy is not tainted by any public or private data release that may fall within the window, we collected data on the release of major economic indicators from www.tradingeconomics.com and verified their timestamp. We found 11 instances where the timestamp was within the identification window; and, hence, we abstained from using the associated tweets in our empirical analysis. The final sample consists of 177 tweets for Canada and 180 for Mexico. As a convention, all timestamps are on Greenwich Mean Time (GMT).

At the time there was some discussion in the economic press as to whether the president could unilaterally withdraw the U.S. from NAFTA and what tariff regime would take place in that case. Article 2205 of NAFTA states that “a Party may withdraw from this Agreement six months after it provides written notice of withdrawal to the other Parties,” so clearly the U.S. could have withdrawn unilaterally from NAFTA. As to whether the president could have initiated the process without Congress approval, Murrill (2016) notes in a legal opinion for the non-partisan Congressional Research Service (CRS) that the president possesses exclusive constitutional authority to communicate with foreign powers and, if characterized under that authority, the president’s delivery of a notice of withdrawal to other free trade agreement (FTA) partners “appears sufficient to terminate the agreements as a matter of international law.” For most trade agreements, such a notice initiates a six-month period until the actual termination of, or withdrawal of a party from, the agreement. Under the Vienna Convention, withdrawing under the provisions of the agreement would have released the U.S. from its obligations under NAFTA from the date that withdrawal or termination became effective (Murrill 2016, p. 10).

Murrill suggests that Section 125 of the Trade Act of 1974, which Congress has made applicable to most free trade agreements signed by the U.S., authorizes the president to proclaim the restoration of tariff rates to what they would be without the agreement. The president shall recommend to Congress the appropriate rates of duty for affected imports within 60 days after termination of, or withdrawal from, an agreement. It is reasonable to assume that President Trump would have proposed tariffs no lower than those under NAFTA for a wide array of Canadian and

⁷ There is no foreign exchange trade during the weekends but otherwise markets are open 24 hours a day in different parts of the world, from 22:00 GMT on Sunday (Sydney) until 22:00 GMT Friday (New York).



Mexican goods entering the U.S. Hufbauer (2017) argues that the president could have invoked the call for “reciprocal and mutually advantageous concessions” in the North American Free Trade Agreement Implementation Act of 1993 to raise U.S. tariffs applicable to Canada, Mexico, or both to the most-favored-nation (MFN) levels that apply to countries that have no free trade agreement with the U.S. Without any presidential action, the section allows the same duties or other import restrictions under the FTA to remain in place for up to a year.⁸

Compared with Mexico, the case of Canada is more complicated because the Canada–U.S. Free Trade Agreement (CUSFTA) was suspended, rather than terminated, when NAFTA entered into force in 1994. Thus, should NAFTA be terminated, the CUSFTA may ensure continuation of free trade between Canada and the United States unless separately terminated. However, the discussion above suggests that the president could unilaterally withdraw the U.S. from CUSFTA as well. Still, the immediate consequences of the U.S. withdrawing from NAFTA would be larger for Mexico than for Canada, and hence, we would expect the exchange rate effects of presidential tweets to be quantitatively larger for the former than for the latter.

Even without leaving NAFTA, the U.S. president was allowed to impose tariffs on Canadian and Mexican steel and aluminum starting in June 2018 under Section 232 of the Trade Expansion Act of 1962. This section permits the president to impose tariffs when “an article is being imported into the United States in such quantities or under such circumstances as to threaten or impair the national security.” The section was upheld by the Supreme Court in 1976, and the Court refused to hear a new challenge to its constitutionality in June 2019. The president also instructed the Secretary of Commerce to investigate the effect of imports of automobiles, trucks, and automotive parts on national security under Section 232 in May 2018.⁹

This discussion shows that presidential tweets criticizing NAFTA or threatening to impose new tariffs, to withdraw the U.S. from NAFTA, or to end the negotiations for a free-trade agreement were potentially consequential and unambiguously signaled the possible increase in tariffs on Canadian and Mexican goods. This discussion also motivates our coding strategy whereby the information content of each tweet is classified into one of five categories, $x \in X = \{1, 2, 3, 4, 5\}$ depending on whether the tweet concerns a possible tariff increase ($x = 1$), the continuation of the status quo or potentially a decrease in tariffs ($x = 2$), border issues ($x = 3$), economic and political issues only marginally related to trade ($x = 4$), or issues unrelated to trade, immigration, or politics ($x = 5$).¹⁰

⁸ This is not to say that Congress is powerless. The Constitution grants Congress the power to impose tariffs on imports, and the presidential power to proclaim modifications to tariff rates derives from a statutory delegation of this power. Presumably, Congress could alter the language of the legislation or repeal the relevant statutes, but this may require bipartisan support.

⁹ Section 232 was also used to justify tariffs on solar panels and washing machines in January 2018, on steel and aluminum in March 2018, and on products made of steel and aluminum (e.g., staples, nails, car bumpers, etc.) in January 2020. These tariffs targeted countries other than Canada and Mexico, although, as discussed in the text, tariffs on steel and aluminum were extended to both countries in June 2018.

¹⁰ The complete list of tweets with timestamps and assigned codes is available from the corresponding author’s web site: www.sites.google.com/site/frugemurcia/welcome/replication-files.



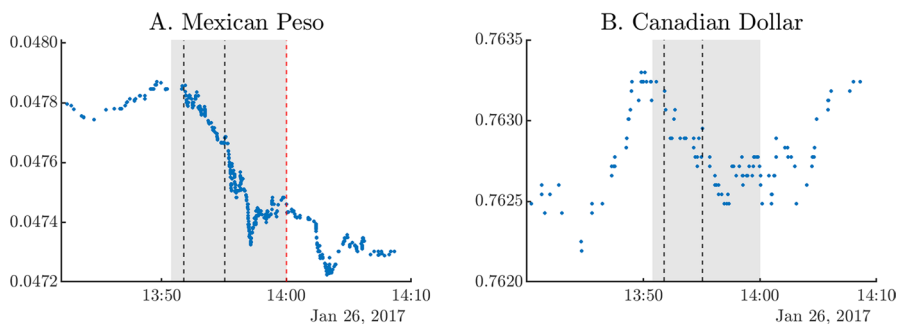


Fig. 1 The figure plots the price of the foreign currency in U.S. dollars around the time a tweet was published at 13:19 GMT

The exchange rate data were obtained from Refinitiv (formerly part of Thompson Reuters) and consist of U.S. dollar/Canadian dollar and U.S. dollar/Mexican peso spot and forward rates. The data are sourced from Refinitiv FX Matching platform, which is a primary inter-dealer electronic trading venue based on the central limit order book, or “CLOB.” We use spot market data at the level of individual executed trades and forward market data on individual quotes for outright forward contracts with tenors ranging from one month to five years. Transaction-level (tick-by-tick) data are essential to implement the high-frequency identification of the effects of presidential tweets on the exchange rates.

3 Case Studies

This section explores the effects of presidential tweets on the U.S. dollar/Canadian dollar and the U.S. dollar/Mexican peso spot exchange rates by focusing on three specific instances.

3.1 Trade Deficit with Mexico (January 26 and 27, 2017)

In the week following his inauguration on January 20, 2017, the U.S. president expressed dissatisfaction with the existing trade arrangement with Mexico as follows:

The United States has a 60 billion dollar trade deficit with Mexico. It has been a one-sided deal from the beginning of NAFTA with massive numbers of jobs and companies lost. If Mexico is unwilling to pay for the badly needed wall then it would be better to cancel the upcoming meeting...

This thread published on January 26 was the first trade-related comment by the president and combined the criticism of NAFTA with immigration issues. Figure 1 plots the U.S. dollar/Mexican peso and U.S. dollar/Canadian dollar exchange rates with the tweet marked in a black dotted line, a data release marked with a red dotted line, and the identification window marked as a shaded grey area. The vertical axis is the



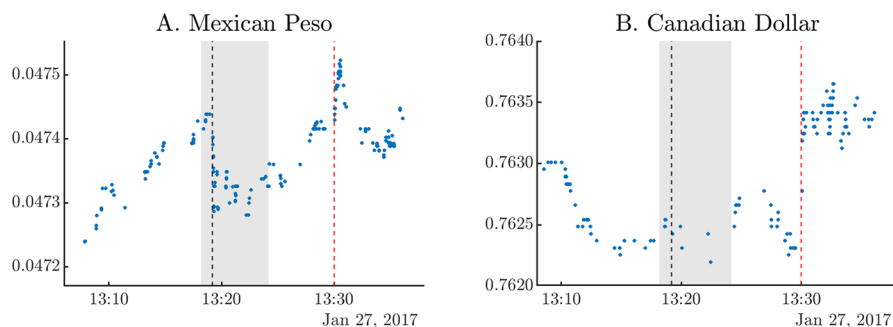


Fig. 2 The figure plots the price of the foreign currency in U.S. dollars around the time a tweet was published at 13:19 GMT

price of the Mexican peso in U.S. dollars (Panel A) and the Canadian dollar in U.S. dollars (Panel B), and the horizontal axis is time. The figure shows an appreciation of the U.S. dollar by 87.1 bps with respect to the Mexican peso. The tweet does not mention Canada explicitly, but the negative depiction of NAFTA led to an appreciation of the U.S. dollar by 8.4 bps with respect to the Canadian dollar.

A follow-up tweet on January 27 leads to an appreciation of the U.S. dollar with respect to the Mexican peso, albeit of a smaller magnitude (see Fig. 2):

Mexico has taken advantage of the U.S. for long enough. Massive trade deficits & little help on the very weak border must change NOW!

The first thread was published shortly before the release of economic data on the Mexican trade balance. One can readily compare their effects and conclude that a U.S. president's tweet on trade can potentially affect the exchange rate by more than the release of actual trade data. Similarly, the follow-up tweet was published not long before the release of data on durable goods orders and the personal consumption expenditures price index in the U.S. One can see that the effects of the tweet and the data release on the U.S. dollar/Mexican peso exchange rate are of comparable magnitude.

3.2 Calls to Renegotiate NAFTA (April 27, 2017)

Approaching the 100th day of his presidency, President Trump was reportedly considering to withdraw the United States from NAFTA. Leaders of both Mexico and Canada reached out to the U.S. president and the withdrawal was averted. These events led to the following tweets:

I received calls from the President of Mexico and the Prime Minister of Canada asking to renegotiate NAFTA rather than terminate. I agreed..

published on April 27, 2017, at 11:12, followed by:



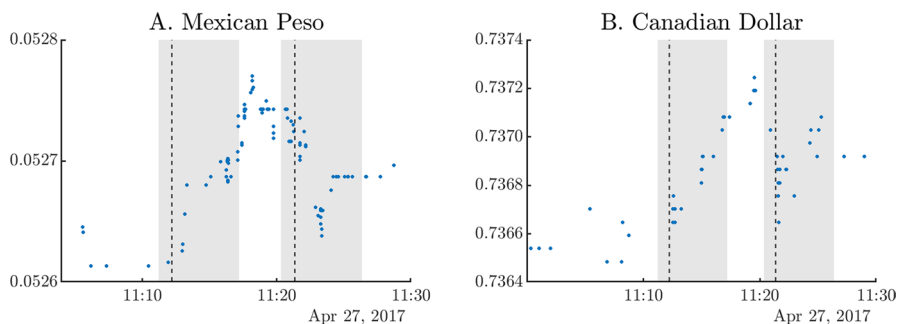


Fig. 3 The figure plots the price of the foreign currency in U.S. dollars around the time tweets were published at 11:12 and 11:21 GMT

...subject to the fact that if we do not reach a fair deal for all, we will then terminate NAFTA. Relationships are good-deal very possible!

published the same day, a few minutes later, at 11:21. The corresponding exchange rate dynamics are displayed in Fig. 3. The first tweet signals that the focus has shifted toward reworking NAFTA. The U.S. exchange rate depreciated by 19.2 bps and 6.6 bps with vis-à-vis to the Mexican peso and Canadian dollar, respectively. The follow-up tweet contains a qualifying statement that maintains withdrawal as a contingency and results in a partial reversal of exchange rates, with the U.S. dollar appreciating by 7.9 bps with respect to the Mexican peso and 3.7 bps with respect to the Canadian dollar.

3.3 Tariffs and Migration Crisis (May 30, 2019)

As attempts to secure funding for the wall along the border with Mexico had proven unsuccessful, President Trump promised to impose escalating tariffs on Mexican goods over illegal immigration. The announcement was made on Twitter on May 30, 2019 at 23:30 as follows:

On June 10th, the United States will impose a 5% Tariff on all goods coming into our Country from Mexico, until such time as illegal migrants coming through Mexico, and into our Country, STOP. The Tariff will gradually increase until the Illegal Immigration problem is remedied, at which time the Tariffs will be removed. Details from the White House to follow.

Figure 4 shows that the U.S. dollar/Mexican peso and U.S. dollar/Canadian exchange rates reacted to this announcement. The U.S. dollar appreciated vis-à-vis the Mexican peso by 61.5 bps and the adjustment persisted beyond our identification time window. Even though the announcement did not directly concern Canada, it put the recently negotiated USMCA trade agreement at risk and led to an appreciation of the U.S. dollar with respect to the Canadian dollar by 5.9 bps.



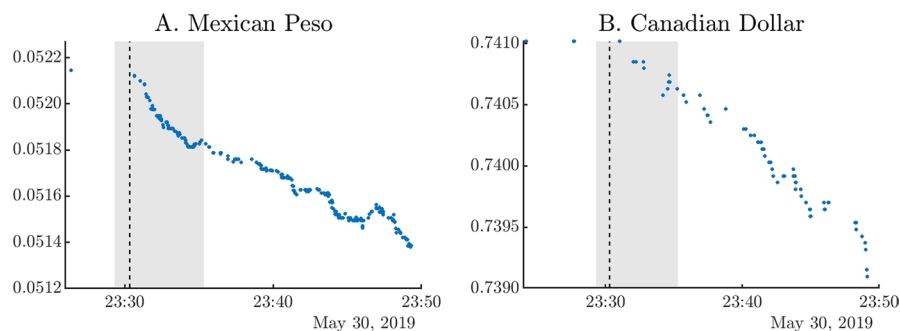


Fig. 4 The figure plots the price of the foreign currency in U.S. dollars around the time a tweet was published at 23:30 GMT

These case studies suggest that the possibility of the increase in trade tariffs leads to the appreciation of the local currency. In the following section, we examine whether this result holds more broadly for the complete set of tweets in our sample.

4 Empirical Analysis

This section reports estimates of the effect of presidential tweets on the U.S. dollar/Canadian dollar and U.S. dollar/Mexican peso exchange rates in an identification window around the tweet timestamps and in the days following their publication. The robustness of the results is discussed with details relegated to an appendix.

4.1 Results

Consider the projection

$$\ln(e_{t+\delta}) - \ln(e_{t-\zeta}) = \alpha + \beta T_t + Z_t' \gamma + u_t, \quad (1)$$

where $\ln(e_{t+\delta}) - \ln(e_{t-\zeta})$ is the change in the nominal bilateral spot exchange rate in a window of $\delta + \zeta$ minutes around the Twitter feed; α is an intercept; β is a coefficient; T_t is a dummy variable that takes value 1 if $x = 1$ (tweet signals a possible increase in tariffs), -1 if $x = 2$ (tweet signals a possible decrease in tariffs or the continuation of the status quo), and 0 otherwise; γ is a vector of coefficients; Z_t is a vector of controls; and u_t is a disturbance independent of T_t and Z_t . Controls are the additional tweets that concern only border issues, and economic and political issues only marginally related to trade. These controls take the form of two dummy variables: the first dummy takes value 1 if $x = 3$ and zero otherwise, and the second dummy takes value 1 if $x = 4$ and zero otherwise. (We exclude the category $x = 5$ as a control because the regression includes an intercept.) Under our identification strategy, the Twitter feed is the only source of new information arriving during the window and ordinary least squares (OLS) delivers a consistent estimate of β , that is,



of the effect of presidential tweets on the exchange rate. Note that the identification window need not be symmetric around the Twitter feed, and in this paper $\varsigma = 1$ minute and $\delta = 5$ minutes.

Benchmark results are reported in Columns 1 and 2 of Table 1 for the U.S. dollar/Canadian dollar and U.S. dollar/Mexican peso exchange rates, respectively. A tweet by the U.S. President signaling a potential increase in tariffs on Canadian goods leads to a 2.61 bps appreciation of the U.S. dollar with respect to the Canadian dollar in the five minutes following its publication. The effect is larger in the case of the U.S. dollar/Mexican peso exchange rate where such a tweet leads to a 4.56 bps appreciation of the U.S. dollar. Both effects are statistically significant at the five percent level. The coefficients of the other tweets are quantitatively smaller and statistically insignificant.¹¹ Thus, the information content of tweets regarding tariffs is different from that of tweets concerning immigration or U.S. politics, and only the former tweets have explanatory power over changes in bilateral exchange rates.

To put these figures in perspective, we compare our estimates with the effects of surprises in macroeconomic data releases on the U.S. dollar/Euro and U.S./Pound exchange rates described by Faust et al. (2007). For example, our estimated effects are roughly comparable to the effect of a U.S. non-farm payrolls release with 40,000 jobs above expectation or a U.S. trade balance higher than expected by half a billion dollars. Table 2 reports the mean, median, and standard deviation of the changes (in absolute value) for each exchange rate in all six-minute time intervals of the full sample from January 1, 2016, to November 29, 2019, and compares them with the estimated effect of a presidential tweet threatening a tariff increase. Since the distribution is positively skewed, the mean of the distribution is larger than the median. The table shows that the effect of a tweet is 1.79 times the median change in the U.S. dollar/Canadian dollar and 4.34 times the median change in the U.S. dollar/Mexican peso exchange rates and comparable in magnitude to one standard deviation of the distribution. Hence, we conclude that the quantitative effect of a presidential tweet is not only statistically significant, but economically significant as well.

The fact that the effect is larger on the U.S. dollar/Mexican peso than on the U.S. dollar/Canadian dollar exchange rate may be due to the fact that Canada could have continued to trade freely with the U.S. after NAFTA's termination under the previous Canada-U.S. Free Trade Agreement, which was suspended, but not terminated, in 1994. Steinberg (2020) finds that the aggregate effects of NAFTA's termination would be larger for Mexico than for Canada. For instance, the decrease in consumption would be 0.26% in Mexico versus 0.13% in Canada, and the decrease in investment would be 0.66% in Mexico versus 0.56% in Canada (see Table 5 in Steinberg, 2020, p. 846). The decrease in welfare in the short- and long-run would be larger in Mexico than in Canada as well (see Table 6 in Steinberg, 2020, p. 850). While our estimates concern the nominal exchange rate, it is safe to assume that the price level is unchanged over the time window around the tweet. Hence, the

¹¹ This explains the result that the F test of the joint significance of all three tweets categories has a p -value above 0.05 in the case of Mexico. A similar result is reported for some of the robustness regressions in Table 4 of Appendix A.



high-frequency movement in the nominal exchange rate translates one-to-one into a high-frequency movement in the real exchange rate in the same direction.

Appendix A examines the robustness of our results to changes in the benchmark specification and reports the results of placebo tests. The appendix shows that conclusions are not driven by the liquidity in different markets, that pre-election tweets have no effect on bilateral exchanges rates,¹² and that results are robust to controlling for data releases in the 60 minutes before or after the identification window, to controlling for trends in the exchange rate that may have been present in the data prior to the presidential tweet, and to using a larger identification window of 15 minutes. Placebo tests show that the effect of presidential tweets on the exchange rate is due to their content about U.S. commercial policy and not to an extraneous reason (for example, who the author of the tweet was).

As an alternative econometric strategy to evaluate the effect of presidential tweets and as robustness check, we perform a difference-in-differences regression using all tweets issued by Donald Trump from January 1, 2016, to November 29, 2019. Of the more than 40,000 tweets in the repository www.thetrumparchive.com, 17,070 were issued in this period and, as mentioned above, 336 of them concerned Canada, Mexico, or their trade agreement. The treatment group is associated with the tweets that signal a possible increase in tariffs on Canadian and Mexican goods selected from these 336 tweets. (These are the tweets coded as $x = 1$ above.) The controls group is associated with the remaining 16,734 tweets, but sample is reduced to 5,171 for Canada and 4,412 for Mexico after excluding retweets (e.g., from Fox News personalities), tweets issued at times when the foreign exchange rate market was closed, and tweets for which there was a public data release during the identification window.¹³ The estimated average treatment effect of the treated (ATT) are -1.95 bps for Canada and -4.23 bps for Mexico. Standard errors are (0.62) and (2.29) , respectively, and were computed using a block bootstrap with 999 replications. The estimated ATTs are statistically significant and their magnitude is similar to the benchmark estimates of -2.61 and -4.56 computed by OLS.

Overall, these results provide empirical support for the conjecture that in a flexible exchange rate regime, an increase in tariffs on foreign goods leads to the appreciation of the local currency (in this case, the U.S. dollar) as, for example, in the theoretical models in Mundell (1961), Krugman (1982), and Barattieri et al. (2021). This exchange rate appreciation may mitigate the expenditure-switching from foreign to local goods intended by the protectionist policy. Finally, our results are in line with recent work by Furceri et al. (2019) and Jeanne and Son (2020). Furceri et al. construct a tariff series based on trade tariff rate data at the product level for 16 sectors in 39 countries and compute the response of several variables to an increase in tariffs. Their VAR results show that an increase of one standard deviation in the

¹² This finding underscores the importance of government communication (tweets by a U.S. president or president elect) over private communication (tweets by a presidential candidate) in the foreign exchange rate market. More broadly, however, private communication that reveals private information may be an important driver of exchange rate fluctuations (see Gholampour and van Wincoop 2019).

¹³ There are instances in which tweets concerning Canada, Mexico, or NAFTA were issued within less than six minutes of unrelated tweets, and vice-versa. For this reason we also eliminated all threads from the sample, but results are very similar if we keep these observations in the sample.



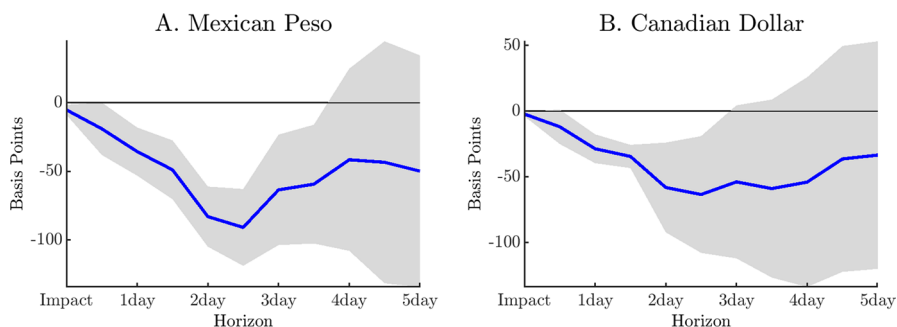


Fig. 5 The figure plots local-projection estimates of the effect of tweets on spot and at various horizons and the associated 90% confidence intervals

tariff rate (about 3.6 percentage points) leads to the real appreciation of the local currency in the short run. Jeanne and Son find that the imposition of tariffs on Chinese goods leads to the appreciation of the U.S. dollar by 0.2% after five hours of the announcement. This appreciation may offset the potential benefits of the tariff increase leading to a small effect on the trade balance and a decrease in output and productivity.¹⁴

4.2 Dynamics

In this section, we examine the dynamic response of the exchange rate to presidential tweets concerning possible tariff increases. We follow a local projections approach by estimating (1) separately for horizons where δ expands with a 12-hour increment for up to 5 days after the tweet.¹⁵ This specification allows us to examine changes in exchange rates cumulated from $t - \zeta$ to $t + \delta$ as in Stock and Watson (2018). Given the volatility of exchange rates, $e_{t+\delta}$ is measured as the average level of transactions over the 12 hours of trading. Differently from other official means of economics-related communication, tweets are not scheduled and do not have a fixed frequency of releases. As a result, the longer is projection horizon δ , the more likely it is that it overlaps with other tweets. In order to limit the temporal contamination of the effects of individual tweets, the sample is restricted by removing tweets that lead or lag other tweets from the first two categories ($x = 1$ and $x = 2$) by less than the longest projection horizon, that is, 5 days. The sample size for the U.S. dollar/Canadian dollar and U.S. dollar/Mexican peso exchange rates regressions is 66 and 68 tweets, respectively.

¹⁴ An earlier paper by Ostry and Rose (1992) uses tariff revenues divided by the value of dutiable imports as a measure of the tariff rate and finds that one cannot reject the hypothesis that tariffs are statistically insignificant determinants of the real exchange rate, output, and the trade balance for the U.S.

¹⁵ Time that spans weekends when the foreign exchange markets are closed is excluded so that if a time horizon includes a weekend, then the horizon is lengthened to accumulate 12 hours of trade time. The local projection estimation was performed using the empirical macro toolbox developed by Canova and Ferroni (2021).



When estimating (1) at multiple horizons δ , we index the regression coefficients, β , by the corresponding horizon and report them in Fig. 5. These coefficients show the dynamic response of the bilateral exchange rate to a presidential tweet concerning possible tariff increases on Canadian and Mexican goods. The impact response in Fig. 5 is based on the same time window as in the benchmark results. The shaded areas represent 90 percent confidence intervals based on the Newey–West HAC-consistent standard errors.

The key conclusion from this analysis is that a presidential tweet concerning possible tariff increases have a persistent effect on foreign exchange rate markets that lasts for up to 5 days and is statistically significant for up to 3 days. In this sense, government communication in the form of tweets can have an effect as long lasting as, for instance, that of press conferences after FOMC meetings. For example, Gorodnichenko et al. (2023) find that the voice tone of the Q & A session after FOMC meetings have an effect on several financial indicators for a few days after the press conference, and Mira Godinho (2022) find that communication measures from FOMC statements affect financial markets for up to between 1 and fifteen days depending on the variable.

5 Interpretation

This section uses a simple general equilibrium model and its non-arbitrage conditions to provide an economic interpretation for the results reported above.

5.1 Theory

The model is a stripped-down version of the two-country model in Obstfeld and Rogoff (1995). The world is populated by identical consumers of which a fraction $n \in (0, 1)$ lives in the home country (h) and the remaining $1 - n$ lives in the foreign country (m). For concreteness, assume that the U.S. is the home country and Canada (or Mexico) is the foreign country. The representative consumer in the home country maximizes

$$E_s \sum_{t=s}^{\infty} \beta^{t-s} U(C_t), \quad (2)$$

where E_s is the expectation conditional on information available at time s , $\beta \in (0, 1)$ is the discount factor, C_t is consumption, and the instantaneous utility function, $U(C_t)$, is strictly increasing, strictly concave, and satisfies the Inada conditions. Consumption is a composite of the home good ($C_{h,t}$) and the foreign good ($C_{m,t}$),

$$C_t = \left((w_h)^{1-\rho} (C_{h,t})^\rho + (w_m)^{1-\rho} (C_{m,t})^\rho \right)^{1/\rho}, \quad (3)$$

where $w_h, w_m \in (0, 1)$ are weights that satisfy the restriction $w_h + w_m = 1$ and $\rho \in (0, 1)$ is a parameter that determines the elasticity of substitution between the



home and foreign goods. Goods are invoiced in terms of the unit of account of the producing country (say, U.S. dollars for home goods and Canadian dollars for foreign goods) but the economy is cashless otherwise. Output in both countries is an exogenous stochastic endowment. The consumer is subject to the budget constraint

$$\begin{aligned} \frac{P_{h,t}C_{h,t}}{P_t} + \frac{e_t(1+\tau_t)P_{m,t}C_{m,t}}{P_t} + \frac{B_{h,t}}{P_t} + \frac{e_tB_{m,t}}{P_t} + \frac{e_tD_{m,t}}{P_t} \\ = \frac{P_{h,t}Y_{h,t}}{P_t} + \frac{i_{t-1}B_{h,t-1}}{P_t} + \frac{e_t i_{t-1}^* B_{m,t-1}}{P_t} \\ + \frac{f_{t-1} i_{t-1}^* D_{m,t-1}}{P_t} + Y_t, \end{aligned}$$

where $P_{h,t}$ is the price of the home good, P_t is the aggregate price index, e_t is the nominal exchange rate in units of local currency per foreign currency, $P_{m,t}$ is the price of the foreign good in foreign currency, τ_t is an *ad-valorem* tariff that is rebated to the consumer as the lump-sum transfer Y_t , $B_{h,t}$ and $B_{m,t}$ denote local and foreign bonds, respectively, D_t is a foreign bond-cum-forward exchange rate contract, f_t is the forward exchange rate, $Y_{h,t}$ is the endowment of the home good, i_{t-1} is the gross home nominal interest rate, and i_{t-1}^* is the gross foreign nominal interest rate. It is assumed that output is known at the beginning of period before the consumer makes consumption and portfolio decisions.

The Euler equations that describe the optimal demand for local and foreign assets are:

$$U'(C_t) = \beta E_t \left(U'(C_{t+1}) \left(\frac{i_t}{\pi_{t+1}} \right) \right), \quad (4)$$

$$U'(C_t) = \beta E_t \left(U'(C_{t+1}) \left(\frac{i_t^*(e_{t+1}/e_t)}{\pi_{t+1}} \right) \right), \quad (5)$$

$$U'(C_t) = \beta E_t \left(U'(C_{t+1}) \left(\frac{i_t^*(f_t/e_t)}{\pi_{t+1}} \right) \right), \quad (6)$$

where π_t is the gross inflation rate. These Euler equations are used below to derive the standard Uncovered Interest Parity and Covered Interest Parity non-arbitrage conditions. The optimal demand for the home and foreign goods is

$$C_{h,t} = w_h \left(\frac{P_{h,t}}{P_t} \right)^{1/(\rho-1)} C_t, \quad (7)$$

$$C_{m,t} = w_m \left(\frac{e_t(1+\tau_t)P_{m,t}}{P_t} \right)^{1/(\rho-1)} C_t, \quad (8)$$

where the aggregate price index is



$$P_t = \left(w_h (P_{h,t})^{\rho/(\rho-1)} + w_m (e_t(1 + \tau_t)P_{m,t})^{\rho/(\rho-1)} \right)^{(\rho-1)/\rho}. \quad (9)$$

In equilibrium, the market for both the home and foreign goods must clear. Denoting foreign variables with a “*” and following Obstfeld and Rogoff (1995) in assuming that the law of one price holds, the equilibrium conditions are

$$\begin{aligned} Y_{h,t} &= n w_h \left(\frac{P_{h,t}}{P_t} \right)^{1/(\rho-1)} C_t + (1-n) w_h^* \left(\frac{P_{h,t}^*/e_t}{P_t^*} \right)^{1/(\rho-1)} C_t^*, \\ Y_{m,t} &= n w_m \left(\frac{e_t(1 + \tau_t)P_{m,t}}{P_t} \right)^{1/(\rho-1)} C_t + (1-n) w_m^* \left(\frac{P_{m,t}^*}{P_t^*} \right)^{1/(\rho-1)} C_t^*. \end{aligned}$$

Define the trade balance in the home country as the difference between exports of the home good and the imports of the foreign good,

$$S_t = (1-n) w_h^* \left(\frac{P_{h,t}^*/e_t}{P_t^*} \right)^{1/(\rho-1)} C_t^* - n w_m \left(\frac{e_t(1 + \tau_t)P_{m,t}}{P_t} \right)^{1/(\rho-1)} C_t. \quad (10)$$

In the short-term (certainly within our six-minute window), it is reasonable to assume that neither consumption nor prices can adjust to an unexpected adjustment in the *ad-valorem* tariff but that financial variables – like the exchange rate in floating regime – traded in liquid markets can instantaneously adjust. The key theoretical result is summarized in the following proposition:

Proposition 1 *In a flexible exchange rate regime, an increase in the tariff on foreign goods leads to the appreciation of the local currency,*

$$\frac{\partial e_t}{\partial \tau_t} < 0.$$

Proof See Appendix B.

Interpreted literally, this result concerns actual, rather than announced or threatened, tariff increases, yet it illustrates the basic mechanism whereby tariffs directly affect the demand of home and foreign goods through relative prices. Since goods are priced in the producer’s currency, relative prices remain unchanged after a tariff increase and the exchange rate appreciation lowers the cost of imports in the home country sufficiently to cancel out the tariff increase. Erceg et al. (2018) derive a similar result in a more general environment with production and price stickiness and, under additional assumptions (e.g., no retaliation), show that the commercial policy is neutral. This result may be interpreted more generally as an example of how a flexible exchange rate regime insulates the local economy from nominal shocks.¹⁶

□

¹⁶ As in Obstfeld and Rogoff (1995), our analysis assumes producer currency pricing (PCP). A literature reviewed in Gopinath and Itskhoki (2022) provides evidence on, and studies implications of, dominant currency pricing (DCP) with the U.S. dollar playing an outsized role in the invoicing of internationally



5.2 Empirics

We now argue that the arbitrage equations of the model suggest three possible channels through which presidential tweets can affect the spot exchange rate, namely the interest rate differential, the future expected exchange rate, and the risk premium, and we empirically explore each channel. Equations (4) and (5) imply an Uncovered Interest Parity (UIP) relation between local and foreign bonds that may be linearly approximated as:

$$\ln(e_t) = E_t \ln(e_{t+k}) + k(i_{k,t}^* - i_{k,t}) - k\rho_{k,t}, \quad (11)$$

where k is the bonds horizon and $\rho_{k,t}$ is the risk premium that U.S. investors would require to hold foreign rather than U.S. k -period bonds, with interest rates and the risk premium expressed on an annualized basis.¹⁷ This specification of UIP is used, for instance, by Faust et al. (2007) to study changes in the exchange rate and interest rates in response to macroeconomic announcements, and we follow their work below. Taking the first difference in (11) around our identification window results in

$$\Delta \ln(e_t) = \Delta E_t \ln(e_{t+k}) + k\Delta(i_{k,t}^* - i_{k,t}) - k\Delta\rho_{k,t}, \quad (12)$$

where Δ is a difference operator that captures the change in the variables in a window of $\delta + \zeta$ minutes around the Twitter feed. Our previous analysis measures the effect of presidential tweets on the left-hand-side of (12), but the right-hand-side of (12) shows that this effect may be associated with changes in the interest rate differential, the expected future exchange rates, and/or the risk premium. We now examine the changes in each of these three components following the tweets.

First, following the literature (for example, Lustig et al. 2011; Du and Schreger 2016; Hassan and Zhang 2021), we construct a measure of interest rate differentials based on spot and forward exchange rates. Specifically, (4) and (6) imply a covered interest parity (CIP) condition that may be linearly approximated as

$$\ln(e_t) = \ln(f_{t+k}) + k(i_{k,t}^* - i_{k,t}). \quad (13)$$

First-differencing (13) around our identification window and rearranging, write the interest-rate differential as

$$k\Delta(i_{k,t}^* - i_{k,t}) = \Delta \ln(e_t) - \Delta \ln(f_{k,t}). \quad (14)$$

Consider now the projection

Footnote 16 (continued)

traded goods. In the context of our analysis, DCP would imply that home-country imports are priced in the home-country currency and their price is, thus, insensitive to exchange rate movements. In turn, this would dampen the trade balance response to a change in the exchange rate. One could then show that an increase in tariffs, as in Proposition 1, would lead to an even stronger appreciation of the home currency than under PCP.

¹⁷ In our model $k = 1$ but it is trivial to incorporate bonds with longer maturities into the model for which UIP must also hold.



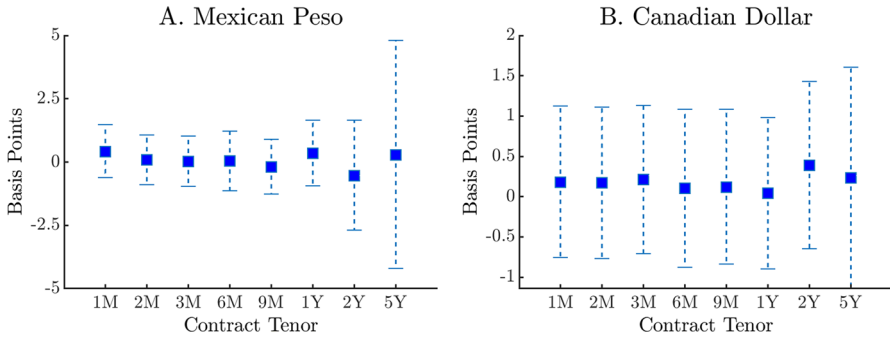


Fig. 6 The figure plots the estimates of the effect of tweets on interest rate differentials at various contract horizons and their associated 95% confidence interval

$$(i_{k,t+\delta}^* - i_{k,t+\delta}) - (i_{k,t-\zeta}^* - i_{k,t-\zeta}) = \alpha + \beta T_t + Z_t' \gamma + u_t,$$

where the interest rate differential is measured by the right-hand side of (14) and the remaining notation is as previously defined.¹⁸ Estimates are reported in Fig. 6. In all cases, the estimate of β is not statistically different from zero. Thus, presidential tweets concerning the possible imposition of tariffs on Canada and Mexico have no explanatory power over the interest rate differential at any contract horizon.

Second, by combining (12) and (14), we see that the forward exchange rate alone captures the two remaining components,

$$\Delta \ln(f_{k,t}) = \Delta E_t \ln(e_{t+k}) - k \Delta \rho_{k,t}. \quad (15)$$

We then estimate the effect of the presidential tweets on $\Delta \ln(f_{k,t})$. That is, we consider the projection

$$\ln(f_{k,t+\delta}) - \ln(f_{k,t-\zeta}) = \alpha + \beta T_t + Z_t' \gamma + u_t,$$

where $\ln(f_{k,t+\delta}) - \ln(f_{k,t-\zeta})$ is the change in the forward exchange rate in the window around the Twitter feed, k is the contract horizon, and the remaining notation is as previously defined. The contract horizons are 1-month, 2-months, 3-months, 6-months, 9-months, 1-year, 2-years, and 5-years ahead. Estimates for Canada and Mexico are reported in Fig. 7, which plots the estimates of β (filled squares) and their 95% confidence intervals for all contracts for both countries.

In the case of the U.S. dollar/Canadian dollar, results using forward rates are quantitatively similar to those obtained using the spot rates for all contracts, even as far as 5-years ahead. For example, the forward exchange rate decreases (i.e., the U.S. dollar appreciates) by 2.79, 2.65, 2.84 bps in the 1-month-, 1-year- and 5-year-forward markets, compared with the 2.61 bps appreciation of the U.S.

¹⁸ Notice that this projection implicitly allows for 1) systematic deviations from CIP, which would be captured by the intercept term, and 2) transitory and potentially heteroskedastic deviations from CIP, which would be captured by the disturbance term.



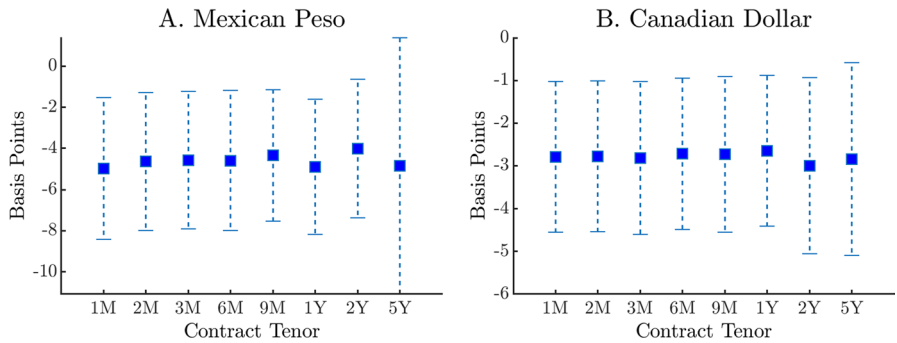


Fig. 7 The figure plots the estimates of the effect of tweets on spot and forward exchange rates at various horizons and their associated 95% confidence interval

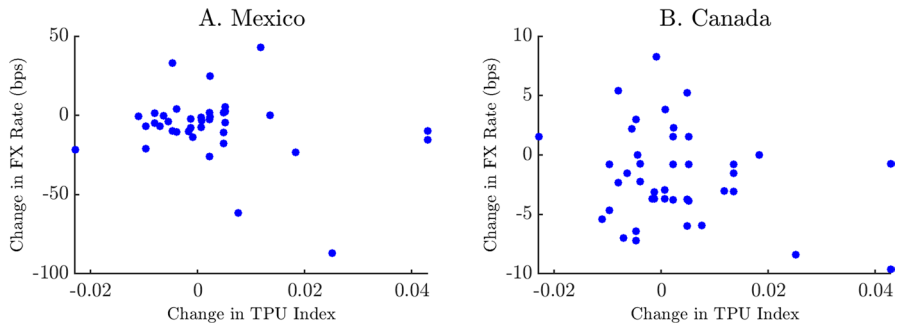


Fig. 8 The figure plots the relation between the change in the daily trade policy uncertainty index Caldara (2020) and the change in the exchange rate

dollar in the spot market. The figure shows that the U.S. dollar appreciation in the forward market due to the presidential tweets is statistically significant in all cases. In the case of U.S. dollar/Mexican peso, Fig. 7 shows that the effect of tweets on the forward rates is comparable in magnitude to the appreciation in the spot market (4.56 bps). These estimates are overall larger than for the U.S. dollar/Canadian dollar exchange rate and significant at standard levels at all horizons, except for 5-year-ahead contracts.

Finally, announcements made via tweets could increase trade policy uncertainty and affect the risk premium. Evaluating this possibility directly in our high-frequency identification scheme is not feasible because data on trade policy uncertainty are not available at the same high frequency as exchange rate trading data. Caldara et al. (2021) construct a daily index of trade policy uncertainty (TPU) based on newspaper coverage of related news and we explore whether there is a statistical correlation between the change in the spot exchange rate in our identification window associated with the tweets from the first category (that is, $x = 1$), which is our variable of interest, and the change in the TPU index. Results are plotted in Fig. 8 for both countries and for changes in the index between the



Table 1 Benchmark results

Description	Exchange rate	
	USD/CAD	USD/MXN
Intercept	0.59 (0.74)	0.72 (1.28)
Tariffs	−2.61* (0.86)	−4.56* (1.82)
Wall	−0.79 (0.84)	1.47 (4.47)
Political	−0.46 (0.89)	0.43 (1.66)
Number obs.	177	180
<i>F</i> -statistic	7.33	0.95
<i>p</i> -value	< 0.01	0.42
<i>R</i> ²	0.11	0.02

The dependent variable is the change in basis points in the U.S. dollar/Canadian dollar (first column) and the U.S. dollar/Mexican peso (second column) spot exchange rate in a window from one minute before to five minutes after the tweet timestamp. The figures in parenthesis are Huber–White heteroskedasticity consistent standard errors. The superscript * denotes statistical significance at the five-percent level

day of the tweet and the day after.¹⁹ The correlation between both variables is basically zero. While we cannot make statements concerning causality, this figure and the results reported above suggest that the protectionist policy signaled by the presidential tweets leads to an U.S. dollar appreciation that is expected to persist into the future by the foreign exchange market.

In summary, results in this section suggest that the effect of presidential tweets on the spot exchange rate is primarily driven by adjustments in the expected future exchange rate. This suggests, in turn, that the foreign exchange markets expected the effect of these tweets, and of the protectionist policy that they implied, to be persistent.

6 Policy Implications

Countries occasionally engage in a commercial policy whereby tariffs and other restrictions on international trade attempt to improve the balance of trade and increase aggregate output. A recent example of such policy was the “America First” policy in the United States during the presidency of Donald Trump. Whether such policies are ultimately successful depend in part on whether tariffs change relative prices and induce a switching in expenditure from foreign to local goods. A key

¹⁹ The plot of the change in the index between the day before and the day of the tweet is reported in the working paper version of this article and supports the same conclusion.



Table 2 Statistics

Description	Exchange rate	
	USD/CAD	USD/MXN
Mean	1.90	2.96
Median	1.46	1.05
Standard deviation	2.66	4.92
Estimated effect of tweets	2.61	4.56

The table reports the mean, median, and standard deviation of the changes (in absolute value) for each exchange rate in all six-minute time intervals of the full sample from January 1, 2016, to November 29, 2019, and compares them with the estimated effect of a presidential tweet threatening a tariff increase

issue, however, concerns the behavior of the exchange rate following the increase in tariffs. This paper uses government communication in the form of tweets issued by the U.S. President concerning actual and possible tariffs on Canadian and Mexican goods, and a clean identification strategy, to measure the effect of trade restrictions on foreign exchange markets. We find that the anticipation of trade restrictions leads to the appreciation of the U.S. dollar by 2.6 bps and 4.6 bps vis-à-vis the Canadian dollar and Mexican peso, respectively, within five minutes of the tweet, and this effect is statistically significant for up to 3 days after the tweet. This result is important because the appreciation of the local currency (in this case the U.S. dollar) would limit the effect of tariffs on relative prices and mitigate the expenditure-switching intended by the protectionist policy. In combination with the related literature mentioned above, our results suggest that commercial policy is likely to have a limited effect on the trade balance in a flexible exchange rate regime.

Appendix A: Robustness and Placebo Tests

Tables 3 and 4 examine the robustness of our results to changes in the benchmark regression. Columns 1 through 8 reports results for the following specifications: Column 1 uses only data from the two most liquid exchange rate markets, namely New York and London; Column 2 considers separately tweets involving only the other NAFTA partner (i.e., Mexico in the case of Canada and vice-versa); Column 3 considers separately tweets before and after Donald Trump’s election as president; Columns 4 and 5 control for instances where there was a public or private data release in the 60 minutes before or after the identification window; Columns 6 and 7 report the result from regressions that, respectively, include the change in the exchange rate in the six minutes and sixty minutes prior to our six-minute identification window; and Column 8 considers the case where the identification window is enlarged to 15 minutes.

Results show that point estimates of the effect of presidential tweets concerning tariffs are quantitatively similar to those reported in Table 1 for the benchmark regression and support the following conclusions. First, results do not appear to be



Table 3 USD/CAD Robustness results and placebo test

Description	Regression								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Intercept	0.93 (0.88)	0.60 (0.74)	0.54 (0.74)	0.72 (0.75)	0.74 (0.72)	0.62 (0.71)	0.60 (0.75)	1.28 (1.07)	-0.39 (0.28)
Tariffs	-3.03* (1.03)	-2.54* (0.97)	-2.71* (0.87)	-2.56* (0.87)	-2.61* (0.86)	-2.55* (0.82)	-2.61* (0.86)	-3.15* 1.40	
Wall	-1.79 (0.99)	-0.79 (0.83)	-0.73 (0.84)	-0.75 (0.85)	-0.81 (0.83)	-0.90 (0.80)	-0.92 (0.84)	-2.32 (1.29)	
Political	-1.15 (1.05)	-0.47 (0.89)	-0.41 (0.89)	-0.37 (0.91)	-0.41 (0.90)	-0.43 (0.86)	-0.46 (0.89)	-1.07 (1.23)	
Mexico		-2.76* (1.01)							
Pre-election			-1.70 (1.67)						
Data before				-0.60 (0.63)					
Data after					-0.45 (0.60)				
Lag						-0.21* (0.10)	-0.01 (0.03)		
Unrelated									-0.06 (1.72)
Number obs.	127	177	177	177	177	176	176	155	177
F-statistic	6.97	5.48	5.63	5.77	5.65	7.20	5.56	3.17	< 0.01
p-value	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	< 0.01	0.03	0.96
R ²	0.15	0.11	0.12	0.12	0.12	0.14	0.12	0.06	< 0.01

The dependent variable is the change in basis points in the U.S. dollar/Canadian dollar spot exchange rate in a window from one minute before to five minutes after the tweet timestamp. The figures in parenthesis are Huber–White heteroskedasticity consistent standard errors. The superscript * denotes statistical significance at the five percent level

driven by the lower liquidity in the other markets in our sample. Second, in the case of Canada, tweets mentioning Mexico alone have the same effect on the U.S. dollar/Canadian dollar exchange rate as tweets mentioning both countries, NAFTA, or Canada alone, but in the case of Mexico, tweets mentioning only Canada have basically no effect on the U.S. dollar/Mexican peso exchange rate. Possible explanations for this result are that the number of tweets mentioning only Canada is smaller (11) than the number mentioning only Mexico (19) and that the language in the latter tweets is considerably more aggressive than that in the former and more likely to receive the attention of traders. Third, pre-election tweets have no effect on exchanges rates, suggesting that results are driven by tweets following the election on November 8, 2016. Fourth, results are robust to controlling for trends in the exchange rate that may have been present in the data prior to the presidential tweet. The reason is that



Table 4 USD/MXN Robustness results and placebo test

Description	Regression								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Intercept	0.11 (1.40)	0.58 (1.34)	0.48 (1.34)	1.87 (1.43)	1.49 (1.59)	0.76 (1.31)	0.38 (1.32)	1.20 (2.20)	0.20 (1.65)
Tariffs	−3.94* (1.83)	−5.62* (2.12)	−5.05* (2.00)	−4.40* (1.85)	−4.62* (1.82)	−4.53* (1.85)	−4.54* (1.83)	−5.53* (2.62)	
Wall	−3.70* (1.70)	1.61 (4.50)	1.70 (4.50)	1.62 (4.55)	1.36 (4.41)	−2.60 (1.55)	−2.51 (1.55)	0.68 (2.59)	
Political	0.45 (1.71)	0.57 (1.71)	0.67 (1.70)	1.19 (1.82)	0.81 (1.68)	0.38 (1.66)	1.04 (1.72)	−4.86 (3.34)	
Canada		0.51 (2.88)							
Pre-election			−0.29 (1.77)						
Data before				−4.43 (2.65)					
Data after					−2.41 (2.28)				
Lag						−0.15 (0.14)	0.07 (0.05)		
Unrelated									1.62 (2.33)
Number obs.	130	180	180	180	180	179	179	157	180
F-statistic	3.63	0.90	0.79	1.13	0.84	3.12	3.85	2.20	0.03
p-value	0.01	0.47	0.53	0.34	0.50	0.02	0.01	0.09	0.86
R ²	0.08	0.02	0.02	0.03	0.02	0.07	0.08	0.04	< 0.01

The dependent variable is the change in basis points in the U.S. dollar/Mexican peso spot exchange rate in a window from one minute before to five minutes after the tweet timestamp. The figures in parenthesis are Huber–White heteroskedasticity consistent standard errors. The superscript * denotes statistical significance at the five percent level

– as can be seen in the estimates in Columns 6 and 7 – exchange rate changes are only mildly serially correlated at this high frequency. Finally, results are robust to using a larger window.

We carried out placebo tests to verify that the effect of presidential tweets on the exchange rate is indeed the result of their content about the U.S. commercial policy and not of an extraneous reason. Placebo tests repeat the analysis using a different part of the dataset where no intervention occurred and ask whether the conjectured effect on the exchange rate is present or not. The natural placebo are tweets by the U.S. president that concern Canada and Mexico but are unrelated to trade, immigration, or U.S. politics. As described in Sect. 2, these tweets were coded as $x = 5$ and so the dummy variable T_t in (1) takes value 1 if $x = 5$ and 0 otherwise. Given the uninformative nature of these tweets from the policy perspective, one would



not expect them to have a systematic effect on the bilateral exchange rate, and evidence of the contrary would call into question our interpretation of the results above. Indeed, results are reported in Column 9 of Tables 3 and 4 show that the coefficient of these presidential tweets is not statistically significant at standard levels for either the U.S. dollar/Canadian dollar or the U.S. dollar/Mexican peso exchange rate.

Appendix B: Proof of Proposition 1

Using (10), define the implicit function,

$$\Gamma(e_t, \tau_t) = (1 - n)w_h^* \left(\frac{P_{h,t}^*/e_t}{P_t^*} \right)^{1/(\rho-1)} C_t^* - nw_m \left(\frac{e_t(1 + \tau_t)P_{m,t}}{P_t} \right)^{1/(\rho-1)} C_t - S_t.$$

By the implicit function theorem,

$$\frac{\partial e_t}{\partial \tau_t} = - \left(\frac{\partial \Gamma(e_t, \tau_t)/\tau_t}{\partial \Gamma(e_t, \tau_t)/e_t} \right).$$

Consider first the numerator $\partial \Gamma(e_t, \tau_t)/\tau_t$ and note that

$$\begin{aligned} \frac{\partial \Gamma(e_t, \tau_t)}{\partial \tau_t} &= -nw_m C_t \left(\frac{1}{\rho - 1} \right) \left(\frac{e_t(1 + \tau_t)P_{m,t}}{P_t} \right)^{1/(\rho-1)-1} \left(\frac{e_t P_{m,t}}{(P_t)^2} \right) \\ &\quad \left(P_t - (1 + \tau_t) \left(\frac{\partial P_t}{\partial \tau_t} \right) \right). \end{aligned}$$

Using (9), one can show that the last term in the right-hand side is

$$P_t - (1 + \tau_t) \left(\frac{\partial P_t}{\partial \tau_t} \right) = P_t \left(1 - \frac{w_m(e_t(1 + \tau_t)P_{m,t})^{\rho/(\rho-1)}}{w_h(P_{h,t})^{\rho/(\rho-1)} + w_m(e_t(1 + \tau_t)P_{m,t})^{\rho/(\rho-1)}} \right) > 0.$$

Recalling that $\rho \in (0, 1)$ so that $1/(\rho - 1) < 0$, it follows that $\partial \Gamma(e_t, \tau_t)/\partial \tau_t > 0$. Consider now the denominator $\partial \Gamma(e_t, \tau_t)/e_t$ and note that

$$\begin{aligned} \frac{\partial \Gamma(e_t, \tau_t)}{\partial e_t} &= (1 - n)w_h^* C_t^* \left(\frac{-1}{\rho - 1} \right) \left(\frac{P_{h,t}^*/e_t}{P_t^*} \right)^{1/(\rho-1)-1} \frac{P_{h,t}^*}{(e_t P_t^*)^2} \left(P_t^* + e_t \left(\frac{\partial P_t^*}{\partial e_t} \right) \right) \\ &\quad + nw_m C_t \left(\frac{-1}{\rho - 1} \right) \left(\frac{e_t(1 + \tau_t)P_{m,t}}{P_t} \right)^{1/(\rho-1)-1} \left(\frac{(1 + \tau_t)P_{m,t}}{(P_t)^2} \right) \left(P_t - e_t \left(\frac{\partial P_t}{\partial e_t} \right) \right). \end{aligned}$$

As before, one can show that

$$P_t^* + e_t \left(\frac{\partial P_t^*}{\partial e_t} \right) = P_t^* \left(1 - \frac{w_h^*(P_{h,t}^*/e_t)^{\rho/(\rho-1)}}{w_h^*(P_{h,t}^*/e_t)^{\rho/(\rho-1)} + w_m^*(P_{m,t}^*)^{\rho/(\rho-1)}} \right) > 0,$$



and

$$P_t - e_t \left(\frac{\partial P_t}{\partial e_t} \right) = P_t \left(1 - \frac{w_m (e_t (1 + \tau_t) P_{m,t})^{\rho/(\rho-1)}}{w_h (P_{h,t})^{\rho/(\rho-1)} + w_m (e_t (1 + \tau_t) P_{m,t})^{\rho/(\rho-1)}} \right) > 0.$$

which implies that $\partial \Gamma(e_t, \tau_t) / \partial e_t > 0$. Since $\partial \Gamma(e_t, \tau_t) / \partial \tau_t > 0$ and $\partial \Gamma(e_t, \tau_t) / \partial e_t$ are both positive, it follows that

$$\frac{\partial e_t}{\partial \tau_t} < 0. \quad \square$$

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